

John “J.D.” Squire

506 S Mathews Ave, Urbana, IL 61801

(210) 649-7458 | jdsquire@outlook.com | jsquire4@illinois.edu | <https://john-squire.github.io>

I am a fourth-year medical student with a passion for engineering and medicine. By harnessing the power of technology today and of the future, I believe we can improve the quality of life for everyone around the globe.

Education

Doctor of Medicine Carle Illinois College of Medicine, University of Illinois at Urbana-Champaign, Urbana, IL	May 2026
Bachelor of Science in Biomedical Engineering, <i>summa cum laude</i> The University of Texas at Dallas, Richardson, TX	May 2022

Research Experience

Graduate Researcher <i>Magnetic Resonance Functional Imaging Laboratory, University of Illinois at Urbana-Champaign</i> Developed machine learning methods to extract information about the brain in magnetic resonance elastography data	Dec 2022 – present
Graduate Researcher <i>Mobility and Fall Prevention Research Laboratory, University of Illinois at Urbana-Champaign</i> Characterized cognitive and motor function in gait with the goal of developing objective and standardized diagnostic measures of motor impairment and treatment	Dec 2022 – Dec 2024
Undergraduate Researcher <i>Molecular Imaging and Optical Nanotherapeutics Laboratory, University of Texas at Dallas</i> Characterized various nanoscale drugs for head and neck cancers under light-activated, x-ray activated, and γ-ray activated conditions	Aug 2021 – Jun 2022
Premier College Intern <i>Air Force Research Laboratory, Wright-Patterson Air Force Base</i> - Worked in the 711th Human Performance Wing in the Bioanalytics focusing on toxicology research - Utilized HTML, CSS, JS, PHP, Python, GUI development, and machine learning	May – Aug 2021
Undergraduate Researcher <i>Molecular Radiation Biology Research Laboratory, University of Texas Southwestern Medical Center</i> - A research partnership between UT Dallas and UT Southwestern under the 2021 Green Fellowship - Acquired imaging data in 3D tumor models for head and neck cancer as well as imaged and modeled time-dependent distribution of x-ray responsive nanoparticles in response to varying doses of radiation therapy - Investigated the use of light-activatable nanotechnology as a facilitator for radiation therapy in radiation-resistant tumors of the head and neck	Jan – May 2021
Undergraduate Researcher <i>Systems for Augmenting Human Mechanics Laboratory, University of Texas at Dallas</i> Learned about neural network architecture and building a neural network to identify diseases related to the hip joint	Jan – Dec 2020
Undergraduate Researcher <i>Speech Disorders and Technology Laboratory, University of Texas at Dallas</i> - Clark Summer Research Program participant - Developed MATLAB programs that connected speech data with machine learning model	Jun 2018 – Jun 2019

Honors and Awards

Second Place Pitch, Cozad New Venture Challenge - Awarded \$500 for achieving second place for the Cozad New Venture Challenge Pitch Competition - Pitched in front of investors about the Intra-Fallopian Device, a novel reversible long-term contraception start-up concept for the Health Innovation Track	May 2025
Finalist Presentation, Global Consortium of Innovation and Engineering in Medicine Invited to host an oral and poster presentation titled “MREureka!: Efficiently and Effectively Segmenting Brain Elastographs with nnU-Net” about my research in applying machine learning to magnetic resonance elastography in the brain	Apr 2025

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Air Force Health Professions Scholarship Program

- Four-year scholarship to cover medical school tuition and fees
- Serving in the Air Force Reserve as a 2nd Lieutenant

May 2022 –
present

Carle Illinois College of Medicine Scholarship

- Four-year scholarship covering \$7,000 of educational expenses each year

May 2022 –
present

First Place for Website Design, FBLA-PBL National Leadership Conference

- Awarded \$1,000 for achieving first place for the Website Design challenge
- Achieved first place at the Texas State Leadership Conference to qualify for nationals

Jun 2020

San Antonio Pipeliners Association Scholarship

- Recipient of the 2019 scholarship covering \$4,000 of educational expenses
- Recipient of the 2020 scholarship covering \$5,000 of educational expenses

Jul 2019 –
Jul 2020

First Place for Mobile Application Design, FBLA-PBL National Leadership Conference

- Awarded \$1,000 for achieving first place for the Mobile Application Design challenge
- Achieved first place at the Texas State Leadership Conference to qualify for nationals

Jun 2019

Society of American Military Engineers Scholarship

- Recipient of the 2019 SAME San Antonio Post Scholarship covering \$5,000 of educational expenses
- Recipient of the 2020 SAME San Antonio Post Scholarship covering \$3,000 of educational expenses
- Recipient of the 2020 SAME Fort Worth Post Scholarship covering \$2,500 of educational expenses

May 2019 –
May 2020

Academic Excellence Scholarship

- Four-year scholarship to cover undergraduate tuition and fees at the University of Texas at Dallas

Aug 2018 –
May 2022

The First Tee Scholars Program

- Selected as the First Tee Scholar from the First Tee of Greater San Antonio program, receiving \$2,500 for educational expenses each year for four years

Aug 2018 –
May 2022

Burger King Scholars Program Scholarship

- Recipient of the 2018 Burger King Scholars award covering \$1,000 of educational expenses

Aug 2018

National Honor Society Scholarship

- Selected as a National Semi-Finalist for the 2018 NHS Scholarship Program, receiving \$2,850 for educational expenses

Aug 2018

Publications

Squire JD, Anderson AT, Sutton BP. MREureka!: Impact of reconstruction approach, phase unwrapping, and inversion method on brain MRE data. September 2025. Pre-Print.

Wong A, Tin A, Lin K, **Squire J**, Limaye S, Bosshardt A, Yang J, Nelson C, Vaidyam A. A Novel EHR Training Platform for Modern Medical Education. *University of Illinois Journal of Medicine*. 2025. Accepted.

Finneran M, **Squire J**, Gordhan A, Nardone E. Dural arteriovenous fistula mimicking a stroke: A misdiagnosis of two months. *Radiology Case Reports*. 2024;19(12):5995-5999. <https://doi.org/10.1016/j.radcr.2024.09.043>

Squire JD, Anderson AT, Johnson CL, Sutton BP. ANTs, BET, or...neither? An exploration of brain masking and machine learning tools applied to magnetic resonance elastography. In: *2024 46th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*.; 2024:1-4. <https://doi.org/10.1109/EMBC53108.2024.10781675>

Shah N, **Squire J**, Guirguis M, Saha D, Hoyt K, Wang KK-H, Agarwal V, Obaid G. Deep-Tissue Activation of Photonomedicines: An Update and Clinical Perspectives. *Cancers*. 2022; 14(8):2004. <https://doi.org/10.3390/cancers14082004>

Poster Presentations

Bosshardt A, Bowman C, Brolly G, Byron N, Cheah D, Chen M, Goldman M, Li M, Ling M, Lopez M, Min D, Patel S, Ray J, **Squire J**, Wong R. QI Project Presentations. Poster presented: *Medication Understanding and Reconciliation in a Student Run Clinic*. Urbana, Medication Understanding and Reconciliation in a Student Run Clinic IL, United States of America; 03/31/2025.

Bosshardt A, **Squire JD**, Ackerson E. Southern Radiological Conference. Poster presented: *Radiologic Confirmation of Chilaiiditi Syndrome in an Adult Female*. Point Clear, AL, United States of America; 01/25/2025.

Chun P, **Squire J**. 2023 Cozad New Venture Challenge Demo Day. Poster presented: *Bonobo.care: Altruism Redefined*. Champaign, IL, United States of America; 04/13/2023.

Squire J. 2024 Cozad New Venture Challenge Demo Day. Poster presented: *Tumestent: The minimally-invasive stenting solution for erectile dysfunction*. Champaign, IL, United States of America; 04/11/2024.

Squire J, Morton C. Discovery Learning Presentation Day. Poster presented: *Becoming a Military Clinician*. Champaign, IL, United States of America; 07/28/2023.

Updated: 11/07/2025

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- Squire J**, Obaid G, Saha D. Green Fellowship Poster Presentation Day. Poster presented: *Efficacy of Nanoformulation-Enhanced Photodynamic Therapy Excited by Radiation in Head and Neck Cancer Cells*. Dallas, TX, United States of America; 05/07/2021.
- Squire J**, Tummala V, Sequeira A, Bosshardt A, Akram S. 2025 Cozad New Venture Challenge Demo Day. Poster presented: *The Intra-Fallopian Device: The novel reversible form of contraception without the regret of surgical sterilization*. Champaign, IL, United States of America; 04/02/2025.
- Squire J**, Wang J, Anson L, Clark Summer Research Program Poster Presentation Day. Poster presented: *Wavcutter: A Tool for Speech Articulation Data Preprocessing*. Richardson, TX, United States of America; 08/03/2018.
- Squire JD**, Anderson AT, Johnson CL, Sutton BP. 22nd Annual Health Innovaton Research Day. Poster presented: *ANTs, BET, or...neither? An exploration of brain masking and machine learning tools applied to magnetic resonance elastography*. Champaign, IL, United States of America; 04/05/2024.
- Squire JD**, Anderson AT, Johnson CL, Sutton BP. 2024 Annual International Conference of the IEEE Engineering in Medicine and Biology Society. Poster presented: *ANTs, BET, or...neither? An exploration of brain masking and machine learning tools applied to magnetic resonance elastography*. Orlando, FL, United States of America; 07/16/2024.
- Squire JD**, Anderson AT, Sutton BP. 2025 Global Consortium of Innovation and Engineering in Medicine. Poster presented: *MREureka!: Efficiently and Effectively Segmenting Brain Elastographs with nnU-Net*. Champaign, IL, United States of America; 04/06/2025.
- Squire JD**, Najafali D, Janbahan M, Akram S, Vyas A, Mustafa IM. 2025 Global Consortium of Innovation and Engineering in Medicine. Poster presented: *Arthritis to Meningitis: A rare case of rheumatoid meningitis*. Urbana, IL, United States of America; 04/06/2025.
- Squire JD**, Tummala VE, Sequeira A, Bosshardt A, Akram S. 2025 Cozad New Venture Challenge. Poster presented: *The Intra-Fallopian Device: The novel reversible form of contraception without the regret of surgical sterilization*. Champaign, IL, United States of America; 04/02/2025.
- Vaidyam A, Wong T, Bosshardt A, Limaye S, **Squire J**, Lin K, Spade R, Jamieson S, Nelson C. 2nd Annual Medical Education Experience (MEdX) 2024. Poster presented: *A Teaching Electronic Medical Records System for Preclinical Students towards Clinical Simulation and Evaluation*. Champaign, IL, United States of America; 01/25/2024.
- Vaidyam A, Wong T, Bosshardt A, Limaye S, **Squire J**, Lin K, Spade R, Jamieson S, Nelson C. 22nd Annual Health Innovation Research Day. Poster presented: *Student-Driven Engineering of Teaching Electronic Medical Records Systems for Clinical Simulation and Evaluation*. Champaign, IL, United States of America; 04/05/2024.

Oral Presentations

- Najafali D, **Squire JD**, Gayed N, McNeil L, Wallon R. A student-faculty partnership to support preparation for USMLE Step 1. Oral presentation: 3rd Annual Medical Education Experience (MEdX) 2025; 01/30/2025. Champaign, IL, United States of America.
- Squire JD**. Made in China 2025: A Health Perspective. Oral presentation: Officer Training School; 06/23/2023. Montgomery, AL, United States of America.
- Squire JD**. Becoming an Adult: The comprehensive transition solution for pediatric patients with chronic diseases. Oral presentation: IDEA Project Presentations; 04/14/2024. Urbana, IL, United States of America.
- Squire JD**. ScanCheck: The AI-based tool that provides automatic real-time feedback on scan quality. Oral presentation: IDEA Project Presentations; 05/19/2024. Urbana, IL, United States of America.
- Squire JD**. Tumestent: The minimally-invasive stenting solution for erectile dysfunction. Oral presentation: IDEA Project Presentations; 07/28/2024. Urbana, IL, United States of America.
- Squire JD**. De-Escaject: A Novel Device Solution for Aggravated Patients. Oral presentation: IDEA Project Presentations; 11/08/2024. Urbana, IL, United States of America.
- Squire JD**. UltraSense: The AI-powered sweeping fetal anatomy ultrasound. Oral presentation: IDEA Project Presentations; 12/15/2024. Urbana, IL, United States of America.
- Squire JD**. Scoping It Out: A New Workflow and Healthcare Business Model. Oral presentation: IDEA Project Presentations; 03/16/2025. Urbana, IL, United States of America.
- Squire JD**, Anderson AT, Sutton BP. MREureka!: Efficiently and Effectively Segmenting Brain Elastographs with nnU-Net. Oral presentation: 2025 Global Consortium of Innovation and Engineering in Medicine; 04/07/2025. Champaign, IL, United States of America.

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Skills and Proficiencies

- ACLS Provider Certified (Sept 2025 – Sept 2027)
- BLS Provider Certified (May 2024 – May 2026)
- Institute for Healthcare Improvement (IHI) Basic Certificate in Quality and Safety (Sept 2025)
- New Venture Development 2025 (Grainger College of Engineering) (May 2025)
- Basic clinical skills in family medicine, internal medicine, pediatrics, surgery, ob/gyn, neurology, and psychiatry through the Carle Illinois College of Medicine
- Entrepreneurship skills in healthcare innovations and customer discovery through the National Science Foundation (NSF) I-Corps Program and the Cozad New Venture Challenge
- Research skills in cell culture, nanoparticle synthesis, and machine learning through various research projects
- Programming skills in Java, Python, HTML, CSS, PHP, JavaScript, MATLAB, Arduino, and Microsoft PowerApps through self-teaching and research projects
- Engineering skills in SolidWorks CAD and 3D printing through various engineering projects
- Video editing skill in VEGAS Pro 15 through personal projects and presentations

Professional Activities

Treasurer/Marketing Chair, Personal Finances and Investments Interest Group <i>Carle Illinois College of Medicine, University of Illinois at Urbana-Champaign</i> • Served as a founding executive board member, encouraging students to become involved in managing their own finances and how to begin building their wealth beginning from medical school	Dec 2024 – present
Voting Member, Clinical Curriculum Subcommittee <i>Carle Illinois College of Medicine, University of Illinois at Urbana-Champaign</i> • Served as voting member to bring a student perspective of the clinical curriculum and improve it for future classes	Sept 2023 – present
Voting Member, Electives Subcommittee <i>Carle Illinois College of Medicine, University of Illinois at Urbana-Champaign</i> • Served as a voting member to make decisions on proposed permanent and student created classroom, clinical, or research electives	Sept 2023 – present
President, Federal Medical Professionals Interest Group <i>Carle Illinois College of Medicine, University of Illinois at Urbana-Champaign</i> • Served as a founding executive board member in creating a supportive environment for other medical students who plan to serve in federal position after medical school, like HPSP students	Sept 2023 – present
Marketing Chair, CIMED Student Government <i>Carle Illinois College of Medicine, University of Illinois at Urbana-Champaign</i> • Served as a member of the voting body and the main point of contact for marketing affairs	Dec 2023 – Dec 2024
Mentor, CIMED Wellness Mentoring Program <i>Carle Illinois College of Medicine, University of Illinois at Urbana-Champaign</i> • Served as a mentor for incoming first year medical students in their transition to medical school, being a resource for academic engagement, social involvement, community service, and navigating medical school in general	Jun 2023 – Jun 2024
Member, Radiology Interest Group <i>Carle Illinois College of Medicine, University of Illinois at Urbana-Champaign</i> • Engaged in meetings revolving around the practice of radiology	Jun 2022 – present
Vice President of Marketing, Texas Phi Beta Lambda/Future Business Leaders of America <i>Naveen Jindal School of Management, University of Texas at Dallas</i> • Served as an executive voting member and marketing liaison for the Texas region of PBL/FBLA chapters	Aug 2020 – May 2022
Mentor, Freshman Mentor Program <i>Office of Undergraduate Education, University of Texas at Dallas</i> • Served as a mentor for freshman students to help them adjust to college life through academics, social life, and college in general	Aug 2019 – May 2022
Voting Member, Victor L. Worsfold Grant Program Fund Selection Committee <i>Hobson Wildenthal Honors College, University of Texas at Dallas</i> • Selected student-led volunteer initiatives to be funded by the Worsfold Grant each semester	Aug 2018 – May 2022

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Professional Memberships

- American College of Radiology (ACR)
- Association of Military Surgeons of the United States (AMSUS), The Society of Federal Health Professionals
- International Society for Magnetic Resonance in Medicine (ISMRM)
- IEEE Engineering in Medicine and Biology Society (IEEE EMBS)
- Illinois State Medical Society (ISMS)
- Society of American Military Engineers (SAME)
- National Society of Professional Engineers (NSPE)